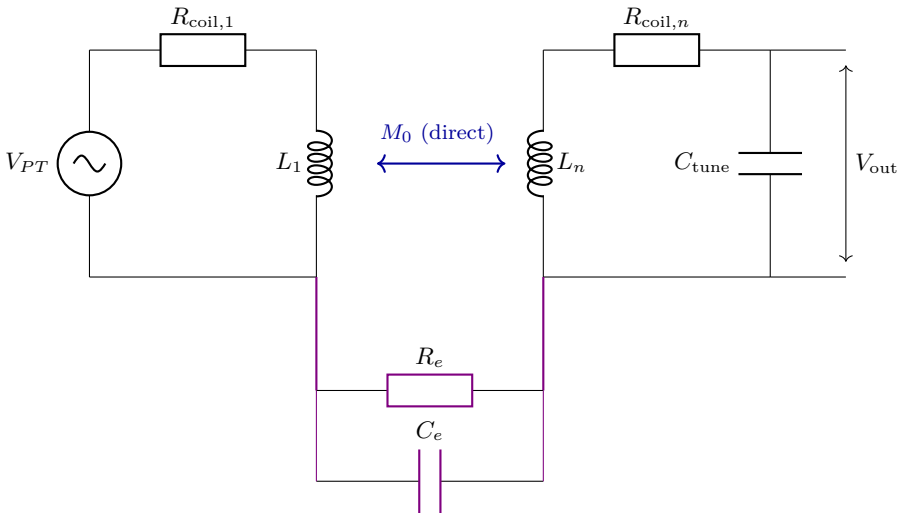


PT generator (primary)

MR receive coil (secondary)



tissue admittivity $\gamma = \sigma + j\omega\epsilon \equiv R_e \parallel C_e$

$R_e \propto 1/\sigma$ (eddy), $C_e \propto \epsilon$ (displacement)

modulated by cardiac motion $\rightarrow M_n = \Delta S_{n,1}/S_{n,1}$

mutual: $Z_{n,1}^{\text{body}} = \iiint \gamma (\mathbf{e}_1 \cdot \mathbf{e}_n) dV$

self-loading: $Z_{nn}^{\text{body}} = \iiint \gamma |\mathbf{e}_n|^2 dV$